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    for successor in get_successors(current_state):
        if successor not in visited:
            visited.add(successor)
            queue.append((successor, path + [successor]))

    return None

if __name__ == '__main__':
    solution = solve_missionaries_cannibals()

    if solution:
        for step_num, step in enumerate(solution):
            m, c, b = step
            boat_pos = "Left" if b == 1 else "Right"
            print("Step", step_num, "- Left Side: M =", m, "C =", c, "| Boat:", boat_pos, "| Right
Side: M =", 3 - m, "C =", 3 - c)
        else:
            print("No solution found.")

```

OUTPUT:

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Step 0 - Left Side: M = 3 C = 3 | Boat: Left | Right Side: M = 0 C = 0
Step 1 - Left Side: M = 3 C = 1 | Boat: Right | Right Side: M = 0 C = 2
Step 2 - Left Side: M = 3 C = 2 | Boat: Left | Right Side: M = 0 C = 1
Step 3 - Left Side: M = 3 C = 0 | Boat: Right | Right Side: M = 0 C = 3
Step 4 - Left Side: M = 3 C = 1 | Boat: Left | Right Side: M = 0 C = 2
Step 5 - Left Side: M = 1 C = 1 | Boat: Right | Right Side: M = 2 C = 2
Step 6 - Left Side: M = 2 C = 2 | Boat: Left | Right Side: M = 1 C = 1
Step 7 - Left Side: M = 0 C = 2 | Boat: Right | Right Side: M = 3 C = 1
Step 8 - Left Side: M = 0 C = 3 | Boat: Left | Right Side: M = 3 C = 0
Step 9 - Left Side: M = 0 C = 1 | Boat: Right | Right Side: M = 3 C = 2
Step 10 - Left Side: M = 1 C = 1 | Boat: Left | Right Side: M = 2 C = 2
Step 11 - Left Side: M = 0 C = 0 | Boat: Right | Right Side: M = 3 C = 3

```

Result: Hence, The program was executed and the output was found to be correct.